🎓 Agentic AI Fundamentals Quiz

🧠 Section A: Multiple Choice Questions (MCQs)

1. **What is Agentic AI?**
   A. AI that only performs predefined tasks
   B. AI systems that act autonomously to achieve goals
   C. AI used only for data storage
   D. AI that requires constant human supervision

   Answer:B – AI systems that act autonomously to achieve goals

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2. **Which of the following best describes an “agent” in AI?**
   A. A database
   B. A program that perceives and acts upon an environment
   C. A neural network only
   D. A hardware device
   Answer: B – A program that perceives and acts upon an environment

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3. **What is the primary goal of an AI agent?**
   A. To maximize randomness
   B. To achieve a defined objective or goal
   C. To reduce memory usage
   D. To store large datasets
   Answer: **B** – To achieve a defined objective or goal

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4. **Which component is responsible for deciding actions in an agent?**
   A. Sensor
   B. Actuator
   C. Policy/Decision-making module
   D. Memory storage
   Answer: **C** – Policy/Decision-making module

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5. **In Agentic AI, “planning” refers to:**
   A. Collecting data
   B. Creating a sequence of actions to achieve a goal
   C. Training a model
   D. Visualizing results
   Answer: **B** – Creating a sequence of actions to achieve a goal

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6. **Which of the following is NOT a typical agent role?**
   A. Perception
   B. Reasoning
   C. Execution
   D. Compilation
   Answer: **D** – Compilation

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7. Reasoning in AI agents helps to:
A. Store information
B. Make logical decisions based on data
C. Increase hardware speed
D. Reduce dataset size
Answer: **B** – Make logical decisions based on data

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8. What does “execution” mean in Agentic AI?
A. Designing algorithms
B. Running planned actions in the environment
C. Training a model
D. Debugging code
Answer: **B** – Running planned actions in the environment

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9. Which type of agent uses condition-action rules?
A. Learning agent
B. Model-based agent
C. Simple reflex agent
D. Utility-based agent
Answer: **C** – Simple reflex agent

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10. Multi-agent systems involve:
A. Only one AI system
B. Multiple agents interacting with each other
C. Only human interaction
D. Only hardware devices
Answer: **B** – Multiple agents interacting with each other

✍️ Section B: Short Answer Questions

11. Define Agentic AI in your own words.
Answer: Agentic AI refers to intelligent systems that can independently think, plan, and take actions to achieve specific goals without constant human control.

12. List any three roles of an AI agent.
Answer:
- Perception (understanding the environment)
- Reasoning (making decisions)
- Execution (performing actions)

13. What is the difference between planning and execution?
Answer: Planning is deciding what actions to take to achieve a goal, while execution is actually performing those actions in the real world.

14. Explain the importance of reasoning in AI agents.
Answer: Reasoning helps AI agents analyze information, understand situations, and make logical decisions instead of acting randomly.

15. What are sensors and actuators in an agent system?
Answer: Sensors collect information from the environment (inputs), while actuators perform actions based on decisions (outputs).

🔍 Section C: Application-Based Questions

16.
A delivery robot must pick up packages and deliver them efficiently.

- Identify the **agent roles** involved.

  Answer: Perception: Detects roads, obstacles, and packages

- Planning: Chooses the best route
- Execution: Moves and delivers packages
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17.
An AI chatbot analyzes user queries and responds intelligently.

- Explain how **reasoning** is used.

  Answer :The chatbot uses reasoning to understand the user's query, analyze the context, and generate an appropriate and meaningful response.

18.
Design a simple agent for a **smart home system**.

- What would be its inputs, outputs, and goals?

  Answer: **Inputs:** Temperature sensors, motion sensors, light sensors

- **Outputs:** Control of lights, AC, fan, alarms
- **Goals:** Improve comfort, save energy, ensure safety
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19.
Explain how **planning + reasoning + execution** work together in a self-driving car.

Answer: Planning: Decides the route and path

 Reasoning: Understands traffic rules and detects obstacles

 Execution: Controls steering, braking, and acceleration

20.
Why is autonomy important in Agentic AI? Give a real-world example.
Answer: Autonomy allows AI to operate independently without human intervention, making systems faster and more efficient.

Example: A self-driving car can make real-time driving decisions without human input.

⚙️ Section D: Higher-Order Thinking

21. Compare Agentic AI with traditional rule-based systems.
Answer: Agentic AI: Learns, adapts, and makes dynamic decisions

 Rule-Based Systems: Follow fixed rules and cannot adapt

- **22.** What challenges might arise in multi-agent systems?
  Answer: Communication issues between agents
- Conflicts in decision-making
- Difficulty in coordination

- **23.** How can Agentic AI improve real-world decision-making?
  Answer: Makes faster decisions
- Analyzes large amounts of data
- Reduces human errors

24. Explain how feedback helps improve agent performance.
Answer: Feedback helps the agent learn from past actions, improve accuracy, and make better decisions in the future.

25. Can Agentic AI function without human input? Justify your answer.
Answer: Agentic AI can work independently to some extent, but human input is still needed for initial setup, monitoring, and handling complex situations.